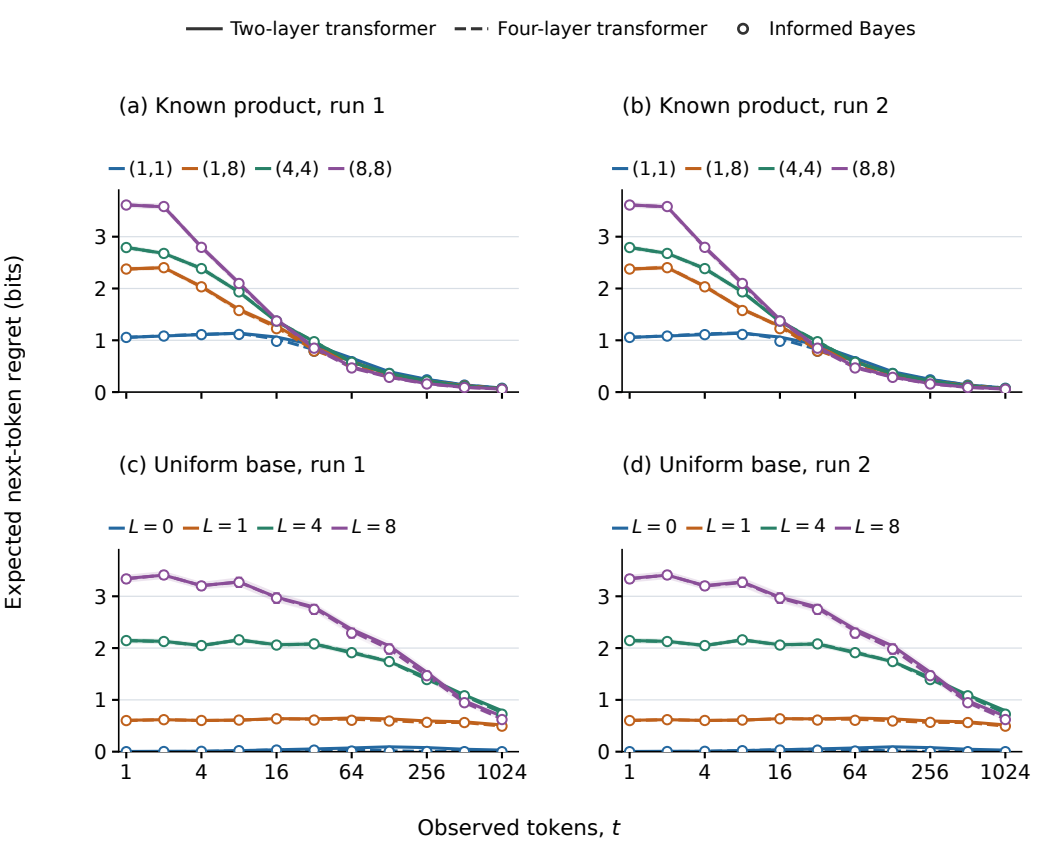
Two-layer transformer
Four-layer transformer
Informed Bayes
(a) Known product, run 1
(b) Known product, run 2
(c) Uniform base, run 1
(d) Uniform base, run 2
(1,1)
(1,8)
(4,4)
(8,8)
$L = 0$
$L = 1$
$L = 4$
$L = 8$
Expected next-token regret (bits)
Observed tokens, t
0
1
2
3
1
4
16
64
256
1024